

AI-ML Assignment 1

Medical Insurance Cost Prediction using Multiple Linear Regression

Name: Rathlavath Karthik

Registration Number: 23BCE11601

Application Number: IN26010749

Batch Number: 2(B)

✓ Task 1: Data Understanding

```
from google.colab import files
import pandas as pd

uploaded = files.upload()
```

insurance.csv

insurance.csv(text/csv) - 55628 bytes, last modified: 7/21/2026 - 100% done
Saving insurance.csv to insurance.csv

✓ Import Libraries

```
# Import required libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
# Load Dataset
```

```
df = pd.read_csv("insurance.csv")

print("Dataset Loaded Successfully!")
```

Dataset Loaded Successfully!

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges	
0	19	female	27.900	0	yes	southwest	16884.92400	
1	18	male	33.770	1	no	southeast	1725.55230	
2	28	male	33.000	3	no	southeast	4449.46200	
3	33	male	22.705	0	no	northwest	21984.47061	
4	32	male	28.880	0	no	northwest	3866.85520	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0   age         1338 non-null   int64  
 1   sex         1338 non-null   object  
 2   bmi         1338 non-null   float64 
 3   children    1338 non-null   int64  
 4   smoker      1338 non-null   object  
 5   region      1338 non-null   object  
 6   charges     1338 non-null   float64 
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
print("Rows and Columns:")
print(df.shape)
```

```
Rows and Columns:
(1338, 7)
```

```
numerical_features = df.select_dtypes(include=['int64','float64']).columns

print("Numerical Features:")
print(numerical_features)
```

```
Numerical Features:
Index(['age', 'bmi', 'children', 'charges'], dtype='object')
```

```
categorical_features = df.select_dtypes(include=['object']).columns

print("Categorical Features:")
print(categorical_features)
```

```
Categorical Features:
Index(['sex', 'smoker', 'region'], dtype='object')
```

```
target_variable = "charges"
```

```
print("Target Variable:")
print(target_variable)
```

```
Target Variable:
charges
```

```
print("Missing Values")
```

```
df.isnull().sum()
```

Missing Values

	0
age	0
sex	0
bmi	0
children	0
smoker	0
region	0
charges	0

dtype: int64

Task 2: Data Preprocessing

```
encoder = LabelEncoder()
```

```
df["sex"] = encoder.fit_transform(df["sex"])
df["smoker"] = encoder.fit_transform(df["smoker"])
df["region"] = encoder.fit_transform(df["region"])
```

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520

Next steps:

[Generate code with df](#)
[New interactive sheet](#)

```
X = df[['age', 'sex', 'bmi', 'children', 'smoker', 'region']]
```

```
y = df['charges']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print("Training Data Shape :", X_train.shape)
print("Testing Data Shape :", X_test.shape)
```

```
Training Data Shape : (1070, 6)
Testing Data Shape : (268, 6)
```

```
model = LinearRegression()

model.fit(X_train, y_train)

print("Model Trained Successfully!")
```

Model Trained Successfully!

```
y_pred = model.predict(X_test)

print("Prediction Completed!")
```

Prediction Completed!

✓ Task 3: Model Development

```
comparison = pd.DataFrame({
    "Actual Charges": y_test,
    "Predicted Charges": y_pred
})
```

```
comparison.head(10)
```

	Actual Charges	Predicted Charges
764	9095.06825	8924.407244
887	5272.17580	7116.295018
890	29330.98315	36909.013521
1293	9301.89355	9507.874691
259	33750.29180	27013.350008
1312	4536.25900	10790.779562
899	2117.33885	226.298446
752	14210.53595	16942.715999
1286	3732.62510	1056.630794
707	10264.44210	11267.919973



Next steps:

[Generate code with comparison](#)

[New interactive sheet](#)

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Task 4: Model Evaluation

Task 4: Model Evaluation

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

```
print("Model Evaluation")
```

```
print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("R2 Score:", r2)
```

```
Model Evaluation
Mean Absolute Error (MAE): 4186.508898366433
Mean Squared Error (MSE): 33635210.431178406
R2 Score: 0.7833463107364539
```

Observation 1 The model achieved an R^2 score of approximately 0.78, indicating that it explains around 78% of the variation in insurance charges.

Observation 2 The scatter plot shows that most predicted values are close to the actual values, although some higher insurance charges are not predicted as accurately.

Observation 3 Smoking status has the greatest impact on insurance charges, while age and BMI also significantly influence the predicted costs.

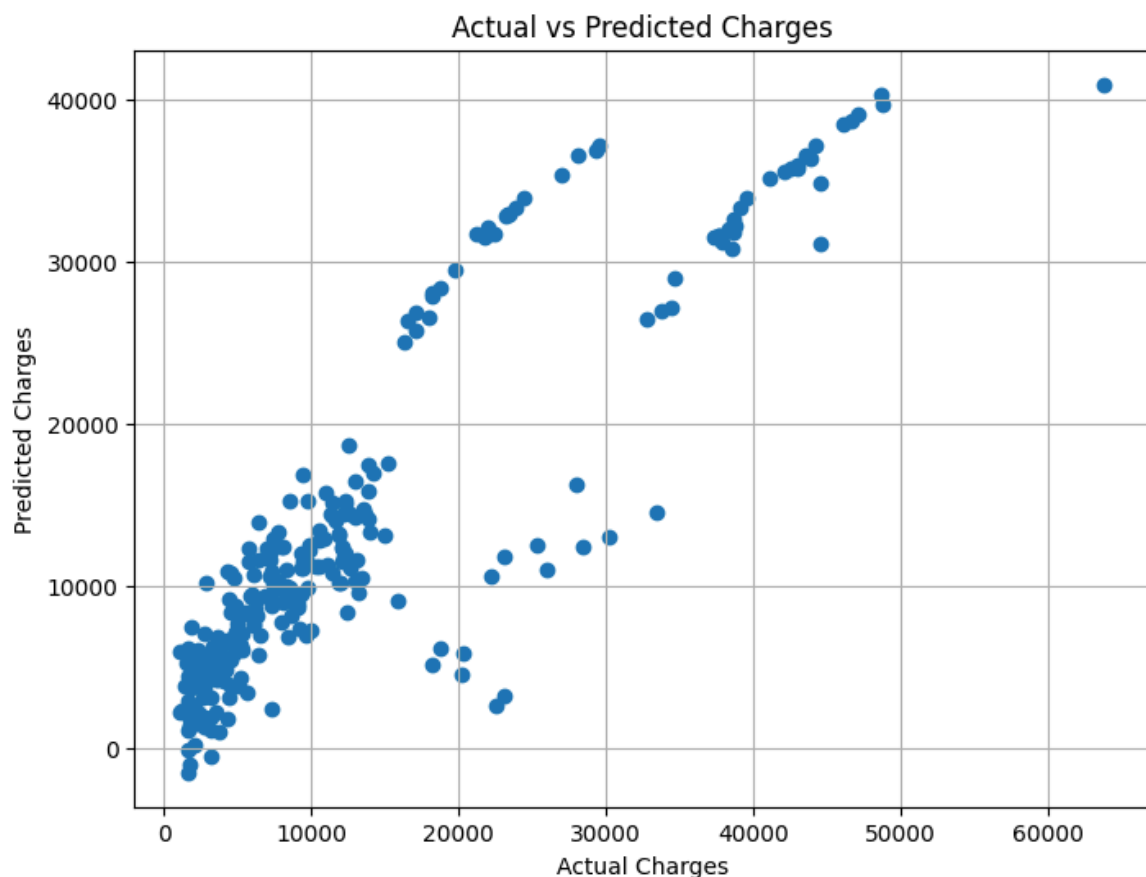
```
plt.figure(figsize=(8,6))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted Charges")

plt.grid(True)

plt.show()
```



```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

coefficients
```

	Feature	Coefficient	
0	age	257.056264	
1	sex	-18.791457	
2	bmi	335.781491	
3	children	425.091456	
4	smoker	23647.818096	
5	region	-271.284266	

Next steps:

[Generate code with coefficients](#)[New interactive sheet](#)

Task 5: Conclusion

The Multiple Linear Regression model was successfully developed to predict medical insurance charges using age, sex, BMI, number of children, smoking status, and region. The model achieved good predictive performance with an MAE of approximately 4186.51, an MSE of 33,635,210.43, and an R^2 score of 0.7833, indicating that it explains about 78% of the variation in insurance charges. Smoking status was found to be the most influential factor affecting insurance costs, followed by BMI and age. One limitation of Linear Regression is that it assumes a linear relationship between the features and the target variable, whereas real-world insurance costs may involve more complex non-linear relationships.

Double-click (or enter) to edit